Date: _____

Question 1 => Pseudocode

```
Start
Print "Enter number of rows :"
Read Rows
Print "Enter number of seats per row:"
Read cols

Print "Enter seating status (0 for available, 1 for booked):"
For i = 0 to rows - 1
    For j = 0 to cols - 1
        Read seats [i] [j]
    End For
End For

Print "Initial seating Arrangement:"
Print "  " (Print column indices 0 to cols - 1)
For i = 0 to rows - 1
    Print "R" + i
    For j = 0 to cols - 1
        If seats [i] [j] == 0 Then
            Print "-"
        Else
            Print "x"
        End If
    End For
End For

total_free = 0
Print "Free seats per row:"
For i = 0 to rows - 1
    row_free = 0
```


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```
For j = 0 to cols - 1
    If seats[i][j] == 0 then
        row_free = row_free + 1
    End If
End For
Print "Row" + i + ": " + row_free
total_free = total_free + row_free
End For
Print "Total free seats: " + total_free

Print "Enter number of customers (K) who want to sit together:"
Read K

If K <= 0 OR K > cols then
    Print "Invalid group size!"
Else
    found = 0
    alloc_row = -1
    alloc_start = -1
    mid = rows / 2

    For d = 0 to rows - 1 AND found == 0
        For i = 0 to 1 AND found == 0
            If i == 0 then
                r = mid - d
            Else
                r = mid + d
            End If

            If (d > 0 OR i == 0) AND (r >= 0 AND r < rows) then
                current_consecutive = 0
```


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```
start_seat = -1

For c = 0 to cols - 1
    If seats [r] [c] == 0 then
        If current_consecutive == 0 then
            start_seat = c
        End If
        current_consecutive = current_consecutive + 1

        If current_consecutive == K then
            alloc_row = r
            alloc_start = start_seat
            found = 1
            Break
        End If
    Else
        current_consecutive = 0
        start_seat = -1
    End If
  End for
 End If
End for
End for

If found == 1 then
    For c = alloc_start to alloc_start + K - 1
        seat [alloc_row] [c] = 1
    End for

    Print "Group of" + K + "seated in Row" + alloc_row + ", Seats" +
        alloc_start + "to" + (alloc_start + K - 1)
Else
```


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Print " Sorry, no consecutive seats available for this group. Seating plan unchanged. "

End If

End If

END.